

MODULE 1 | PART 1
BASIC ALGEBRA
IOQM & RMO 2026

BASIC ALGEBRA

Algebraic Identities

Expansion: Identity $(a + b)^2$

$$(a + b)^2 = a^2 + 2ab + b^2$$

Expansion: Identity $(a - b)^2$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Example:

Real numbers a and b satisfy

$$(a + b)^2 + (a - b)^2 = 100, \quad ab = 6$$

Find $a^4 + b^4$.

Solution:

$$(a + b)^2 + (a - b)^2 = 2(a^2 + b^2) = 100$$

$$\Rightarrow a^2 + b^2 = 50,$$

$$a^4 + b^4 = (a^2 + b^2)^2 - 2(ab)^2$$

$$= 50^2 - 2 \cdot 6^2$$

$$= 2500 - 72$$

$$= 2428$$

Example:

Prove for all real a, b :

$$(a - b)^2 + (a - 2b)^2 + (2a - b)^2 \geq 2ab.$$

Solution:

Expand and collect terms to get

$$6(a^2 + b^2) - 10ab \geq 2ab$$

$$\Rightarrow 6a^2 + 6b^2 - 12ab \geq 0$$

which follows from $(a - b)^2 \geq 0$.

Example:

$$\text{If } (3x - 4)^2 + (4x + 3)^2 = 100,$$

find all real x .

Solution:

$$9x^2 - 24x + 16 + 16x^2 + 24x + 9 = 100$$

$$\Rightarrow 25x^2 + 25 = 100 \Rightarrow x^2 = 3$$

$$\Rightarrow x = \pm\sqrt{3}.$$



Difference of Squares $a^2 - b^2$

$$a^2 - b^2 = (a - b)(a + b)$$

Example:

Simplify $\frac{x^8 - 1}{x^4 - 1}$.

Solution:

$$x^8 - 1 = (x^4 - 1)(x^4 + 1)$$

$$\Rightarrow \frac{x^8 - 1}{x^4 - 1} = x^4 + 1.$$

Example:

Find

$$(1 + 2)(1^2 + 2^2)(1^4 + 2^4) \cdots (1^{2^n} + 2^{2^n})$$

Solution:

Note

$$(1 + 2)(1^2 + 2^2)(1^4 + 2^4) \cdots (1^{2^k} + 2^{2^k}) = \frac{2^{2^{k+1}} - 1}{2^{2^k} - 1}$$

so the product telescopes to $2^{2^{n+1}} - 1$

Cubic Expansions $(a \pm b)^3$

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$



Example:

Compute

$$(3 + \sqrt{5})^3 + (3 - \sqrt{5})^3.$$

Solution:

$$\text{Use } (a + b)^3 + (a - b)^3 = 2a^3 + 6ab^2.$$

Here $a = 3$, $b = \sqrt{5}$:

$$2 \cdot 27 + 6 \cdot 3 \cdot 5 = 54 + 90 = 144.$$

Sum and Difference of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Example:

If $a + b = 1$ and $a^3 + b^3 = 19$, find ab .

Solution:

$$a^3 + b^3 = (a + b)^3 - 3ab(a + b)$$

$$\Rightarrow 19 = 1 - 3ab \Rightarrow ab = -6.$$



Square of a Sum in Three Variables

$$(a + b + c)^2$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Example:

$$\text{If } a + b + c = 6, \quad ab + bc + ca = 11$$

$$\text{find } a^2 + b^2 + c^2.$$

Solution:

$$\begin{aligned}(a + b + c)^2 &= a^2 + b^2 + c^2 \\ &\quad + 2(ab + bc + ca) = 36 \\ \Rightarrow a^2 + b^2 + c^2 &= 36 - 2 \cdot 11 = 14.\end{aligned}$$

The Cubic Symmetry Identity

$$\begin{aligned}a^3 + b^3 + c^3 - 3abc \\ = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)\end{aligned}$$

Special Case: $a + b + c = 0$

Then

$$a^3 + b^3 + c^3 = 3abc$$

Example:

Real x, y, z satisfy

$$x + y + z = 0, \quad x^3 + y^3 + z^3 = 24.$$

Find xyz .

Solution:

By the special case,

$$x^3 + y^3 + z^3 = 3xyz = 24 \Rightarrow xyz = 8.$$

Factorising a Quadratic

$$(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta$$

Example:

Let x, y be roots of $t^2 - 14t + 45 = 0$.

Compute $x^4 + y^4$.

Solution:

$$\begin{aligned}x + y &= 14, \quad xy = 45 \\ \Rightarrow x^2 + y^2 &= 14^2 - 2 \cdot 45 = 196 - 90 \\ &= 106 \\ x^4 + y^4 &= (x^2 + y^2)^2 - 2(xy)^2 \\ &= 106^2 - 2 \cdot 45^2 \\ &= 11236 - 4050 \\ &= 7186.\end{aligned}$$

Example:

Roots x, y of $t^2 - 8t + 7 = 0$.

Compute $x^3 - y^3$.

Solution:

$$\begin{aligned}x + y &= 8, \quad xy = 7. \text{ Then} \\ (x - y)^2 &= 64 - 28 = 36 \Rightarrow x - y = \pm 6, \\ x^2 + xy + y^2 &= 64 - 7 = 57 \\ \Rightarrow x^3 - y^3 &= \pm 6 \cdot 57 = \pm 342.\end{aligned}$$

Factorising a Cubic

$$\begin{aligned}(x - \alpha)(x - \beta)(x - \gamma) &= x^3 - \\ &(\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma\end{aligned}$$

Example:

Factorise $x^3 + 2x^2 - 5x - 6$.

Solution:

Test $x = 2$: $8 + 8 - 10 - 6 = 0$.

So divide by $(x - 2)$:

$$(x^3 + 2x^2 - 5x - 6) \div (x - 2) = x^2 + 4x + 3$$

$$= (x + 1)(x + 3)$$

Hence $(x - 2)(x + 1)(x + 3)$

Factoring $x^4 + x^2 + 1$

$$x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1)$$



1. Three real numbers x, y, z are such that $x^2 + 6y = -17$, $y^2 + 4z = 1$ and $z^2 + 2x = 2$. What is the value of $x^2 + y^2 + z^2$?

[PRMO 2013]



2. Positive integers a and b are such that $a + b = \frac{a}{b} + \frac{b}{a}$. What is the value of $a^2 + b^2$?

[PRMO 2015]



3. Let a, b , and c be real numbers such that $a - 7b + 8c = 4$ and $8a + 4b - c = 7$. What is the value of $a^2 - b^2 + c^2$?

[PRMO 2015]



4. * Let $a = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$, let $b = \frac{x}{z} + \frac{y}{x} + \frac{z}{y}$

and let $c = \left(\frac{x}{y} + \frac{y}{z}\right)\left(\frac{y}{z} + \frac{z}{x}\right)\left(\frac{z}{x} + \frac{x}{y}\right)$.

The value of $|ab - c|$ is:

[IOQM 2024]





5. Find the number of ordered triples (x, y, z) of real numbers that satisfy the system of equations:

$$x + y + z = 7; \quad x^2 + y^2 + z^2 = 27; \quad xyz = 5.$$

[IOQM 2020]



Ratios & Proportions

Basic Ratio Manipulations

❖ A ratio $a : b$ denotes $\frac{a}{b}$.

❖ Equivalent ratios:

$$a : b = c : d \iff \frac{a}{b} = \frac{c}{d}.$$

❖ To use $a : b = m : n$,

$$\text{set } a = mk, \quad b = nk.$$

Example:

$$a : b = 3 : 5, \text{ find } \frac{5a + 3b}{2a - b}.$$

Solution:

Let $a = 3k$, $b = 5k$. Then

$$\frac{5a + 3b}{2a - b} = \frac{5 \cdot 3k + 3 \cdot 5k}{2 \cdot 3k - 5k} = \frac{30k}{k} = 30.$$

Example:

Given $x : y = 7 : 4$, compute $\frac{x^2 - y^2}{x + y}$.

Solution:

Let $x = 7k$, $y = 4k$. Then

$$\frac{x^2 - y^2}{x + y} = \frac{49k^2 - 16k^2}{11k} = \frac{33k^2}{11k} = 3k.$$

$$\text{Equivalently, } \frac{x^2 - y^2}{x + y} = x - y = 3k.$$

Example:

If $m : n = 5 : 2$ and $m + n = 21$, find m and n .

Solution:

Set $m = 5k$, $n = 2k$.

$$\text{Then } 5k + 2k = 21 \Rightarrow k = 3.$$

$$\text{So } m = 15, n = 6.$$



Example:

If $r : s = 4 : 7$, evaluate

$$\frac{r^3 + s^3}{(r + s)(r^2 - rs + s^2)}.$$

Solution:

$$\text{Recall } r^3 + s^3 = (r + s)(r^2 - rs + s^2).$$

$$\text{Thus } \frac{r^3 + s^3}{(r + s)(r^2 - rs + s^2)} = 1.$$

Proportions

- ❖ A proportion is an equality of two ratios:

$$\frac{a}{b} = \frac{c}{d} = k.$$

Then $a = kb$, $c = kd$.

- ❖ For a triple proportion

$$\frac{x_1}{y_1} = \frac{x_2}{y_2} = \frac{x_3}{y_3} = k, \text{ set each } x_i = ky_i.$$

Example:

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$, show that $\frac{a+e}{b+f} = \frac{c+e}{d+f}$.

Solution:

Let the common ratio be k ,

so $a = kb$, $c = kd$, $e = kf$. Then

$$\frac{a+e}{b+f} = \frac{kb+kf}{b+f} = k,$$

$$\frac{c+e}{d+f} = \frac{kd+kf}{d+f} = k.$$

Hence they are equal, each equal to $k = \frac{a}{b}$.

Componendo and Dividendo

For a ratio $\frac{a}{b} = k$:

$$\frac{a+b}{a-b} = \frac{k+1}{k-1}, \quad \frac{a-b}{a+b} = \frac{k-1}{k+1}.$$



Example:

Given $\frac{a}{b} = \frac{5}{2}$, find $\frac{a+b}{a-b}$.

Solution:

Here $k = \frac{a}{b} = \frac{5}{2}$.

Hence

$$\frac{a+b}{a-b} = \frac{k+1}{k-1} = \frac{\frac{5}{2}+1}{\frac{5}{2}-1} = \frac{\frac{7}{2}}{\frac{3}{2}} = \frac{7}{3}.$$

Example:

If $\frac{x-2}{x+2} = 3$, find x using Componendo and Dividendo rule.

Solution:

Set $a = x-2$, $b = x+2$, $k = 3$.

Then $\frac{a+b}{a-b} = \frac{k+1}{k-1} = \frac{4}{2} = 2$

But $a+b = 2x$, $a-b = -4$,

so $\frac{2x}{-4} = 2 \Rightarrow 2x = -8 \Rightarrow x = -4$.

Example:

Given $\frac{p+q}{p-q} = 7$, find $\frac{p}{q}$.

Solution:

Here $k = \frac{p+q}{p-q} = 7$.

By Componendo and Dividendo rule,

$$\frac{p}{q} = \frac{k+1}{k-1} = \frac{7+1}{7-1} = \frac{8}{6} = \frac{4}{3}.$$



Sum of Numerators and Denominators

❖ If $\frac{a}{b} = \frac{c}{d} = k$, then $\frac{a+c}{b+d} = k$

❖ Similarly, $\frac{a-c}{b-d} = k$ when $b \neq d$.

❖ Extends to any number of equal ratios:

$$\frac{a_1}{b_1} = \dots = \frac{a_n}{b_n} = k$$

$$\Rightarrow \frac{a_1 + \dots + a_n}{b_1 + \dots + b_n} = k$$

Example:

If $a : b = c : d = 3 : 4$, find $\frac{a+c}{b+d}$.

Solution:

Let $a = 3x$, $b = 4x$, $c = 3y$, $d = 4y$.

Then $\frac{a+c}{b+d} = \frac{3x+3y}{4x+4y} = \frac{3(x+y)}{4(x+y)} = \frac{3}{4}$.

Compound and Chain Ratios

❖ A compound ratio $a : b$ and $b : c$ gives

$$a : c = \frac{a}{b} \times \frac{b}{c}.$$

❖ For multiple links $a_1 : a_2$, $a_2 : a_3$,

$$a_3 : a_4, \dots, a_{n-1} : a_n,$$

$$a_1 : a_n = \frac{a_1}{a_2} \times \frac{a_2}{a_3} \times \frac{a_3}{a_4} \times \dots \times \frac{a_{n-1}}{a_n}$$

Example:

If $a : b = 2 : 3$ and $b : c = 5 : 7$, find $a : c$

Solution:

$$a : c = \frac{a}{b} \times \frac{b}{c} = \frac{2}{3} \times \frac{5}{7} = \frac{10}{21}$$

Hence $a : c = 10 : 21$

Example:

If $x : y = 4 : 9$, $y : z = 6 : 11$, and

$z : w = 3 : 5$, compute $x : w$.

Solution:

$$x : w = \frac{x}{y} \times \frac{y}{z} \times \frac{z}{w} = \frac{4}{9} \times \frac{6}{11} \times \frac{3}{5}$$

$$= \frac{4 \cdot 6 \cdot 3}{9 \cdot 11 \cdot 5} = \frac{72}{495} = \frac{8}{55}.$$

Hence $x : w = 8 : 55$.

Example:

In $\triangle ABC$, $AB : BC : CA = 3 : 4 : 5$ and the perimeter is 72. Find the side-lengths.

Solution:

Let $AB = 3x$, $BC = 4x$, $CA = 5x$.

Then $3x + 4x + 5x = 12x = 72 \Rightarrow x = 6$.

Hence $AB = 18$, $BC = 24$, $CA = 30$.



6. If $a = b - c$, $b = c - d$, $c = d - a$ and $abcd \neq 0$ then what is the value of

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}?$$

[PRMO 2012]





7. *Let α and β be positive integers such that

$$\frac{16}{37} < \frac{\alpha}{\beta} < \frac{7}{16}.$$

Find the smallest possible value of β .

[IOQM 2023]



Exponents

Fundamental Laws of Exponents

- ❖ $a^m \cdot a^n = a^{m+n}$ (Product rule)
- ❖ $\frac{a^m}{a^n} = a^{m-n}$ (Quotient rule)
- ❖ $(a^m)^n = a^{mn}$ (Power-to-power)
- ❖ $a^0 = 1$ (for $a \neq 0$)
- ❖ $a^{-n} = \frac{1}{a^n}$

Example:

If $2^{x+3} \cdot 2^{2x-1} = 2^8$, find x .

Solution:

By the product rule,

$$2^{(x+3)+(2x-1)} = 2^8 \Rightarrow 2^{3x+2} = 2^8$$

$$\Rightarrow 3x + 2 = 8 \Rightarrow x = 2.$$

Example:

If $\frac{7^{3x}}{7^{x+2}} = 49$, find x .

Solution

By the quotient rule,

$$7^{3x-(x+2)} = 7^2 \Rightarrow 7^{2x-2} = 7^2$$

$$\Rightarrow 2x - 2 = 2 \Rightarrow x = 2.$$

Example:

Simplify and express in the form $x^k y^m$:

$$(x^2 y^{-1})^3 \cdot (x y^2)^{-2}.$$

Solution:

Apply power-to-power and negative exponents:

$$\begin{aligned} & (x^2)^3 (y^{-1})^3 \cdot x^{-2} (y^2)^{-2} \\ &= x^6 y^{-3} \cdot x^{-2} y^{-4} = x^{6-2} y^{-3-4} = x^4 y^{-7} \end{aligned}$$

Hence the result is $x^4 y^{-7}$.

Example:

Evaluate: $3^0 + 2^{-3} + 5^{-1}$.

Solution:

Recall $a^0 = 1$ and $a^{-n} = \frac{1}{a^n}$:

$$\begin{aligned} 3^0 + 2^{-3} + 5^{-1} &= 1 + \frac{1}{8} + \frac{1}{5} \\ &= 1 + 0.125 + 0.2 \\ &= 1.325 \end{aligned}$$



Example:

Solve for x : $\frac{2^{2x} \cdot 4^{x-1}}{8^{x+2}} = \frac{1}{16}$.

Solution:

Express all in base 2:

$$4 = 2^2, 8 = 2^3, \frac{1}{16} = 2^{-4}.$$

$$\begin{aligned} \text{Then } \frac{2^{2x} \cdot (2^2)^{x-1}}{(2^3)^{x+2}} &= \frac{2^{2x} \cdot 2^{2x-2}}{2^{3x+6}} \\ &= 2^{(2x+2x-2)-(3x+6)} \\ &= 2^{x-8} = 2^{-4} \end{aligned}$$

Hence $x - 8 = -4 \Rightarrow x = 4$.

Fractional Exponents and Radicals

❖ $a^{\frac{m}{n}} = \sqrt[n]{a^m}$ (for $a \geq 0$).

❖ $\sqrt[n]{a} = a^{\frac{1}{n}}, \sqrt{a} = a^{\frac{1}{2}}$.

❖ $\left(a^{\frac{m}{n}}\right)^p = a^{\frac{mp}{n}}, (ab)^{\frac{m}{n}} = a^{\frac{m}{n}} b^{\frac{m}{n}}$.

❖ $a^{-\frac{m}{n}} = \frac{1}{a^{\frac{m}{n}}}$

❖ Domain: fractional powers with an even denominator require $a \geq 0$.

Example:

Evaluate $27^{\frac{2}{3}}$.

Solution:

Write $27 = 3^3$,

so $27^{\frac{2}{3}} = (3^3)^{\frac{2}{3}} = 3^{3 \cdot \frac{2}{3}} = 3^2 = 9$.

Example:

Solve for x : $x^{\frac{3}{2}} = 8$.

Solution:

Write $x^{3/2} = (x^{1/2})^3 = 8 = 2^3$.

Thus $x^{1/2} = 2$, giving $x = 4$.

Example:

Simplify $\frac{16^{\frac{3}{4}} \cdot 8^{\frac{2}{3}}}{4^{\frac{5}{2}}}$.

Solution:

Express each base as power of 2:

$$16^{\frac{3}{4}} = (2^4)^{\frac{3}{4}} = 2^3 = 8,$$

$$8^{\frac{2}{3}} = (2^3)^{\frac{2}{3}} = 2^2 = 4,$$

$$4^{\frac{5}{2}} = (2^2)^{\frac{5}{2}} = 2^5 = 32$$

Thus

$$\frac{8 \cdot 4}{32} = \frac{32}{32} = 1.$$



8. Suppose that $4^{X_1} = 5$, $5^{X_2} = 6$,
 $6^{X_3} = 7, \dots, 126^{X_{123}} = 127$, $127^{X_{124}} = 128$.

What is the value of the product

$$X_1 X_2 \dots X_{124}?$$

[PRMO 2012]





9. If $3^x + 2^y = 985$ and $3^x - 2^y = 473$, what is the value of xy ?

[PRMO 2015]



Logarithms

Basic Properties of Logarithms

❖ Definition: $\log_b a = c \iff b^c = a$,
with $a > 0$, $b > 0$, $b \neq 1$.

❖ $\log_b(xy) = \log_b x + \log_b y$.

❖ $\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$.

❖ $\log_b(x^r) = r \log_b x$.

❖ Change-of-base:

$$\log_b a = \frac{\ln a}{\ln b} = \frac{\log_k a}{\log_k b}$$

Example:

Convert to exponential form and find x :

$$\log_3 x = 4.$$

Solution:

Exponential form: $3^4 = x$. Hence $x = 81$.

Example:

Simplify using log-rules: $\log_2(8 \cdot 16)$.

Solution:

$$\log_2 8 + \log_2 16 = 3 + 4 = 7.$$

Example:

Evaluate: $\log_5\left(\frac{125}{25}\right)$.

Solution:

$$\log_5 125 - \log_5 25 = 3 - 2 = 1.$$

Example:

Simplify: $\log_2\left(\frac{2^5 \cdot 8^2}{\sqrt{2}}\right)$.

Solution:

Rewrite in base 2:

$$2^5 \cdot 8^2 = 2^5 \cdot (2^3)^2 = 2^5 \cdot 2^6 = 2^{11},$$

$$\sqrt{2} = 2^{1/2}.$$

So the argument is $\frac{2^{11}}{2^{1/2}} = 2^{11-\frac{1}{2}} = 2^{\frac{21}{2}}$.

$$\text{Thus } \log_2(2^{21/2}) = \frac{21}{2}.$$

Example:

Solve for x : $\log_2 x + \log_2(x - 2) = 3$.

Solution:

Combine logs:

$$\log_2[x(x - 2)] = 3 \implies x(x - 2) = 2^3 = 8$$

$$\implies x^2 - 2x - 8 = 0$$

$$\implies (x - 4)(x + 2) = 0$$

Accept $x = 4$ (since $x > 2$). $x = 4$

Example:

Solve: $2\log_3 x = \log_3(9x - 8)$

Solution:

Left side = $\log_3(x^2)$.

So $\log_3(x^2) = \log_3(9x - 8)$

$$\Rightarrow x^2 = 9x - 8 \Rightarrow x^2 - 9x + 8 = 0$$

$$\Rightarrow (x - 1)(x - 8) = 0$$

Both $x = 1, 8$ satisfy domain $x > \frac{8}{9}$. 1, 8

Example:

Evaluate: $(\log_2 3)(\log_3 4)(\log_4 5)(\log_5 8)$.

Solution:

Use change of base formula:

$$\log_2 3 \cdot \log_3 4 = \log_2 4 = 2,$$

$$\log_4 5 \cdot \log_5 8 = \log_4 8 = \frac{3}{2}.$$

$$\text{Thus the product is } 2 \times \frac{3}{2} = \boxed{3}.$$

Solving Exponential Equations

❖ To solve $a^{f(x)} = b$, take logs:

$$f(x) \ln a = \ln b.$$

❖ Use change-of-base: $\log_a b = \frac{\ln b}{\ln a}$.

Example:

Solve for x : $4^{2x-1} = 8^{x+2}$.

Solution:

Write $4 = 2^2$, $8 = 2^3$:

$$(2^2)^{2x-1} = (2^3)^{x+2} \Rightarrow 2^{4x-2} = 2^{3x+6}$$

$$\Rightarrow 4x - 2 = 3x + 6 \Rightarrow x = 8.$$

Example:

Solve the equation: $5^{2x} - 5^x - 6 = 0$.

Solution:

Set $y = 5^x$. Then $y^2 - y - 6 = 0$ factors:

$$(y - 3)(y + 2) = 0 \Rightarrow y = 3 \text{ or } y = -2$$

Since $5^x > 0$, $5^x = 3$.

$$\text{Hence } x = \log_5 3 = \frac{\ln 3}{\ln 5}.$$

Surds

Basics of Surds

❖ A surd is an expression involving roots that cannot be simplified to a rational number, e.g. $\sqrt{n}$ for non-square n .

❖ **Simplification:**

$$\sqrt{a^2 b} = a\sqrt{b},$$

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

❖ **Like terms:** $a\sqrt{b} + c\sqrt{b} = (a + c)\sqrt{b}$.

❖ **Rationalizing simple denominator:**

$$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}.$$

❖ **Nested radical:** For $a, b > 0$,

$$\sqrt{m + 2\sqrt{n}} = \sqrt{p} + \sqrt{q}$$

$$\text{if } p + q = m, pq = n.$$

Example:

Simplify: $\sqrt{72}$.

Solution:

$$\sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$$

Example:

Simplify: $\sqrt{50} + \sqrt{18}$.

Solution:

$$\sqrt{50} = 5\sqrt{2}, \quad \sqrt{18} = 3\sqrt{2},$$

$$\text{so } 5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$$

Example:

Simplify: $3\sqrt{8} \times 2\sqrt{2}$

Solution:

$$3\sqrt{8} \times 2\sqrt{2} = 6\sqrt{8 \cdot 2} = 6\sqrt{16} = 6 \cdot 4 = 24$$

Example:

Rationalize the denominator: $\frac{1}{2\sqrt{3}}$.

Solution:

Multiply top and bottom by $\sqrt{3}$:

$$\frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{2 \cdot 3} = \frac{\sqrt{3}}{6}.$$

Example:

Simplify the nested radical: $\sqrt{5 + 2\sqrt{6}}$.

Solution:

Seek $p, q > 0$ with

$$(\sqrt{p} + \sqrt{q})^2 = p + q + 2\sqrt{pq} = 5 + 2\sqrt{6}.$$

Thus

$$p + q = 5, \quad pq = 6 \implies (p, q) = (2, 3).$$

Hence

$$\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}$$



Operations with Surds

❖ **Addition/Subtraction:** Combine like surds: $a\sqrt{b} \pm c\sqrt{b} = (a \pm c)\sqrt{b}$

❖ **Multiplication:** Multiply coefficients and radicands: $\sqrt{a}\sqrt{b} = \sqrt{ab}$.

❖ **Division:** $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$.

❖ **Rationalizing Denominators:** Multiply by the conjugate to remove surds:

$$\frac{1}{x \pm \sqrt{y}} \times \frac{x \mp \sqrt{y}}{x \mp \sqrt{y}}.$$

Example:

Simplify: $\sqrt{50} + 2\sqrt{8} - 3\sqrt{2}$

Solution:

$$\sqrt{50} = 5\sqrt{2}, \quad 2\sqrt{8} = 2 \cdot 2\sqrt{2} = 4\sqrt{2}.$$

Hence

$$5\sqrt{2} + 4\sqrt{2} - 3\sqrt{2} = 6\sqrt{2}$$



Example:

Compute: $(\sqrt{3} + \sqrt{7})(\sqrt{3} - \sqrt{7})$.

Solution:

Difference of squares:

$$(\sqrt{3})^2 - (\sqrt{7})^2 = 3 - 7 = -4$$

Example:

Expand and simplify: $(\sqrt{2} + \sqrt{5})^2$

Solution:

$$(\sqrt{2} + \sqrt{5})^2 = 2 + 2\sqrt{10} + 5 = 7 + 2\sqrt{10}.$$

Example:

Simplify: $\frac{\sqrt{6} \cdot \sqrt{15}}{\sqrt{10}}$.

Solution:

$$\frac{\sqrt{90}}{\sqrt{10}} = \sqrt{\frac{90}{10}} = \sqrt{9} = 3$$

Example:

Rationalize the denominator: $\frac{5}{\sqrt{3} + \sqrt{2}}$.

Solution:

Multiply by conjugate:

$$\begin{aligned} \frac{5}{\sqrt{3} + \sqrt{2}} \cdot \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} &= \frac{5(\sqrt{3} - \sqrt{2})}{3 - 2} \\ &= 5(\sqrt{3} - \sqrt{2}). \end{aligned}$$



Solving Equations Involving Surds

- ❖ Isolate one radical term on one side before squaring.
- ❖ Square both sides carefully, expanding cross-terms.
- ❖ Repeat isolation and squaring if multiple radicals remain.
- ❖ Check domain and reject extraneous roots introduced by squaring.
- ❖ Contest tip: Look for symmetry or substitutions to simplify nested radicals.

Example:

Solve: $\sqrt{x+2} + \sqrt{x-3} = 3$

Solution:

Domain: $x - 3 \geq 0 \Rightarrow x \geq 3$.

Isolate one root:

$$\sqrt{x+2} = 3 - \sqrt{x-3}.$$

Square:

$$\begin{aligned} x+2 &= 9 - 6\sqrt{x-3} + (x-3) \\ \Rightarrow 6\sqrt{x-3} &= 4 \Rightarrow \sqrt{x-3} = \frac{2}{3}. \end{aligned}$$

Square again:

$$x-3 = \frac{4}{9} \Rightarrow x = 3 + \frac{4}{9} = \frac{31}{9}.$$

Check $x \geq 3$: OK.

Verify original:

$$\begin{aligned} \sqrt{\frac{31}{9}} + 2 + \sqrt{\frac{31}{9} - 3} &= \sqrt{\frac{49}{9}} + \sqrt{\frac{4}{9}} \\ &= \frac{7}{3} + \frac{2}{3} = 3. \end{aligned}$$

Example:

Solve: $\sqrt{x+5} = x-1$

Solution: $\boxed{4}$.

Domain: $\text{RHS} \geq 0 \Rightarrow x \geq 1$.

Square:

$$x+5 = (x-1)^2 = x^2 - 2x + 1$$

$$\Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0$$

Candidates: $x = 4, -1$.

Domain $x \geq 1$ gives $x = 4$.



10. If $\frac{1}{\sqrt{2011} + \sqrt{2011^2 - 1}} = \sqrt{m} - \sqrt{n}$,
where m and n are positive integers, what is
the value of $m+n$?

[PRMO 2012]



Greatest Integer

The Greatest Integer (Floor) Function

- ❖ Notation: $\lfloor x \rfloor$ is the greatest integer less than or equal to x .
- ❖ If $n \in \mathbb{Z}$ and $n \leq x < n+1$, then $\lfloor x \rfloor = n$.
- ❖ Key property: $\lfloor x+k \rfloor = \lfloor x \rfloor + k$ for integer k .
- ❖ Monotonicity: If $x \leq y$ then $\lfloor x \rfloor \leq \lfloor y \rfloor$.
- ❖ For negative x : e.g. $\lfloor -1.2 \rfloor = -2$.

Example:

Solve: $\lfloor x \rfloor = 2$.

Solution:

By definition, $2 \leq x < 3 \Rightarrow x \in [2, 3)$.

Example:

Solve: $\lfloor x \rfloor = -1$.

Solution:

We require $-1 \leq x < 0$.

Example:

Solve: $\lfloor x+1 \rfloor = 4$.

Solution:

Set $y = x+1$.

Then $\lfloor y \rfloor = 4$ gives $4 \leq y < 5$,

so $4 \leq x+1 < 5 \Rightarrow 3 \leq x < 4$.

Example:Solve: $\lfloor 2x \rfloor = 5$.**Solution:**Let $y = 2x$. Then $5 \leq y < 6$, so

$$5 \leq 2x < 6 \Rightarrow \frac{5}{2} \leq x < 3$$

Example:

$$\left\lfloor \frac{x}{2} \right\rfloor = 3$$

Solution:Set $y = \frac{x}{2}$. Then $3 \leq y < 4$, so

$$3 \leq \frac{x}{2} < 4 \Rightarrow 6 \leq x < 8.$$



11. How many non-negative integral values of x satisfy the equation $\left\lfloor \frac{x}{5} \right\rfloor = \left\lfloor \frac{x}{7} \right\rfloor$?

(Here $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x . For example $\lfloor 3.4 \rfloor = 3$ and $\lfloor -2.3 \rfloor = -3$.)

[PRMO 2012]

Modulus Operator

The Modulus (Absolute Value)

❖ Definition: $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$

❖ Non-negativity: $|x| \geq 0$ for all real x .

❖ Product/Quotient: $|xy| = |x||y|$,

$$\left| \frac{x}{y} \right| = \frac{|x|}{|y|} \text{ for } y \neq 0.$$

❖ Solving $|f(x)| = a$ (with $a \geq 0$) gives $f(x) = a$ or $f(x) = -a$.

Example:Solve: $|x| = 5$.**Solution:**

$|x| = 5$ means $x = 5$ or $x = -5$. $x = 5, -5$.

Example:Solve: $|x - 3| = 7$ **Solution:**

$$x - 3 = 7 \Rightarrow x = 10,$$

$$\text{or } x - 3 = -7$$

$$\Rightarrow x = -4.$$

$$x = 10, -4.$$



Example:

Solve: $|2x + 1| = 9$.

Solution:

$$2x + 1 = 9 \Rightarrow 2x = 8 \Rightarrow x = 4, \text{ or}$$

$$2x + 1 = -9 \Rightarrow 2x = -10 \Rightarrow x = -5.$$

$$x = 4, -5$$

Example:

Solve: $|3x - 6| = 0$.

Solution:

$$|3x - 6| = 0 \text{ implies } 3x - 6 = 0$$

$$\Rightarrow x = 2.$$

**Example:**

Evaluate: $\{3.14\}$.

Solution:

$$\lfloor 3.14 \rfloor = 3, \quad \{3.14\} = 3.14 - 3 = 0.14.$$

Example:

Evaluate: $\{-1.7\}$.

Solution:

$$\lfloor -1.7 \rfloor = -2,$$

$$\{-1.7\} = -1.7 - (-2) = 0.3.$$

Example:

Solve for $x \in [0, 5) : \{x\} = 0.75$.

Solution:

Write $x = n + 0.75$ with $n = \lfloor x \rfloor$

Since $0 \leq n \leq 4$, the solutions are

$$x = 0.75, 1.75, 2.75, 3.75, 4.75.$$

Fractional Part

The Fractional Part Function

- ❖ Notation: $\{x\} = x - \lfloor x \rfloor$.
- ❖ Range: $0 \leq \{x\} < 1$ for all real x .
- ❖ Periodicity: $\{x + n\} = \{x\}$ for any integer n .
- ❖ Relationship: $x = \lfloor x \rfloor + \{x\}$.
- ❖ If $x \in [0, 1)$, then $\{x\} = x$.

Example:

Determine all x such that: $\{x\} = x$.

Solution:

We require $x = \{x\}$ and $0 \leq \{x\} < 1$.

Hence $\lfloor x \rfloor = 0$ and $x \in [0, 1)$. $x \in [0, 1)$

1. (14)

2. (2)

3. (1)

4. (1)

5. (3)

6. $\left(\frac{1}{2}\right)$

7. (23)

8. $\left(\frac{7}{2}\right)$

9. (48)

10. (2011)

11. (9)

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